

Spring 2025

1) a) Information is processed data that is useful so that the decision-maker may take necessary action.

Key qualities of information

1. Availability/Accessibility - Easy to obtain when needed
2. Accuracy - Must be correct and dependable for its purpose.
3. Reliability - Should reflect truth and objectivity
4. Relevance - Must relate to the decision-making purpose.
5. Completeness - Should contain all details required for proper decisions.

b) The software development model used here is the Spiral Model. It divides the project into smaller phases, and in each phase, risk assessment, ~~ph~~ prototyping, and client feedback are performed. This model is particularly useful for high-risk projects with frequently changing requirements. It addresses challenges by:

1. Risk Management - Each phase begins with identifying

and analyzing risks, reducing chances of failure.

2. Iterative Development - The project is developed in repeated cycles (iterations), allowing continuous refinement.

3. Client Feedback - Ensures requirements are updated and aligned with user needs.

4. Flexibility - Adapts easily to changes during the project lifecycle.

Sp 2.4

1 a) System

A group of interrelated components working together to achieve a result.

Stages in building system:

Analysis → Design → Implementation → Maintenance.

System Analyst

Investigates, designs, develops, and maintains information systems.

Roles / Responsibilities

Translate business needs into IT solutions, plan projects, document processes, train users, coordinate with stakeholders.

Qualities

Technical knowledge, communication skills, business understanding, critical thinking.

b) The TechFiesta Mega-sale at TechBudget Zone illustrates the integration of all three levels of information.

* Strategic Information: Top management decides on the long term strategy of offering a 30% discount to boost market share, attract new customers, and strengthen brand image.

* Tactical Information: Middle management plans the sale activities, such as budgeting for the discount, designing marketing campaigns, arranging inventory levels, and scheduling staff between 1st-15th January.

* Operational Information: Day-to-day activities are managed, including processing customer transactions, updating stock, monitoring daily sales, and ensuring smooth checkout and delivery during the sale.

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1) a) * Strategic Information: Showing trendiest feature in app, i.e. calorie count for every item. This is a long-term strategic decision to attract health-conscious customers and gain a competitive advantage.

* Tactical Information: Added in-app features of "top 10 most ordered meals of the week" and "top rated restaurants near you". These are tactical decisions by middle management to improve customer engagement and align with strategic goals.

* Operational Information: In-app pop-up notifications highlighting new features immediately when the user opens the app. This is operational, as it deals with short-term, day-to-day execution to inform customers in real time.

b) Characteristics of Quality Information

Refer sp 25 Q1a

Roles/Responsibilities of System Analyst

- * Analyze user needs to identify engagement challenges
- * Design system features like progress tracking, quizzes, and forums.
- * Translate requirements into technical solutions
- * Coordinate with stakeholders (students, instructors, developers)

- * Ensure usability and effectiveness of features for improving ~~completeness~~ completion rates.

CW

Case 1:

- * Ownership: Customer owns the data; the bank is only custodian.
- * Issue: Using data for marketing without consent is unethical and inappropriate.
- * System Analyst's Role:
 - * Translate business needs into IT solutions.
 - * Ensure ethical use of information
 - * Act as an ethical watchdog in system development
- * Action: James should insist on customer authorization before data is reused.
- * Reason: Protects trust, ensures compliance, avoids legal/reputation risks.

Case 2:

- * Sponsor: Registrar's Office / Administration.
- * Business Need: Improve satisfaction & efficiency in enrollment process

Business Requirements:

- * Anywhere, anytime online enrollment.
- * Real-time course availability updates.
- * Clear instructions & online help.
- * Mobile access for flexibility.

Potential Value:

- * Saves student time, reduces queues.
- * Enhances satisfaction and experience.
- * Improves accuracy and reduces manual errors.
- * Provides reliable enrollment data for decisions.

Constraints/Issues:

- * Limited budget.
- * Need staff & student training.
- * Integration with existing IT system.

Knowledge Management Systems (KMS):

Capture and share organizational knowledge.

Simple and easy to use.

Process is clearly defined.

When requirements are well understood.

2) A system is a set of interrelated components working together to achieve a common goal by taking inputs, processing them, and producing outputs.

Types of Information Systems and their roles:

1) Transaction Processing Systems (TPS):

Handle day-to-day business transactions

2) Management Information Systems (MIS):

Provide summarized reports for decision-making.

3) Decision Support Systems (DSS):

Help managers make non-routine decision using data analysis.

4) Executive Support Systems (ESS):

Assist top management with strategic decisions

5) Knowledge Management Systems (KMS):

Capture and share organizational knowledge.

b) The Systems Development Life Cycle (SDLC) is the structured process of determining how an information system can support business needs, designing it, building it, and delivering it to users. It ensures systematic development, reduces risks, and produces efficient systems.

Key Phases (with e-commerce example):

- 1) Planning: Identify need and feasibility.
- 2) Analysis: Gather requirements.
- 3) Design: Decide system architecture, databases, and interface.
- 4) Implementation & Operation: Code, test, install, and maintain.

2b) For developing an e-commerce platform with small size and no uncertain requirements, the Waterfall Model is suitable.

Advantages:

- * Simple and easy to use.
- * Each phase is clearly defined.
- * Works well when requirements are well understood and fixed.

Disadvantages:

- * Difficult to go back once a phase is completed
- * No working software until late in the cycle.
- * High risk if changes are needed later.

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- 2b) i) For the accounting software, the Waterfall Model is suitable because requirements are clearly defined, stable, and the client expects quick delivery. Waterfall works best for small projects with fixed requirements.
- ii) For the social networking platform, the Agile Model is appropriate since it supports iterative development, adapts to changing needs, and allows frequent releases for user feedback.

Justification:

- * Waterfall: Clear structure, easy to manage, but rigid if requirements change.
- * Agile: Flexible, adaptive, user-focused, but requires skilled teams and strong collaboration.

2) (or) For developing an online learning platform, the Agile Model is most suitable.

Justification:

- * Agile supports incremental and iterative developments allowing new modules and features to be added continuously.
- * Regular feedback from educators can be incorporated quickly, improving usability.
- * It ensures flexibility to adapt to changing learning needs and technology.
- * Advantages: Adaptive, fast delivery, user-focused.
- * Disadvantages: Requires skilled teams and may lack detailed documentation.

b) Methodologies: Provide guidelines for every stage of the SDLC.

Example: Agile, Waterfall.

Models: Graphical or conceptual representations of system components.

Example: Data Flow Diagram, ER Diagram.

Tools: Software applications that support system development.

Example: Microsoft Project, CASE tools

Techniques: Step-by-step methods used to perform tasks.

Example: User interviewing, Software testing.

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3) Factors Affecting System Projects

1) Internal Factors - These come from within the organization, such as:

- * Strategic plan: Guides IT priorities

- * Top managers: Their directives often trigger large projects.

- * User requests: User demand better services or improvements.

- * IT department: Proposes new systems or upgrades.

- * Existing data: Errors or limitations in current systems push new projects.

2) External Factors - These are outside influences,

including:

- * Technology: Rapid changes drive system updates.

- * Suppliers: Integration through systems like EDI and JIT.

- * Customers: CRM systems to meet customer needs.

* Competitors: Competitive pressure to adopt new systems.

* Economy: ~~Econ~~ Economic conditions after system planning.

* Government: Regulations shape system design.

3b) Steps to Conduct Economic Feasibility Analysis

1) Identify Costs and Benefits - Include development costs, operational costs, tangible and intangible benefits.

Example: Development costs = salaries, training;

Tangible benefit = increased sales.

2) Assign Values to Costs and Benefits - Estimate monetary values with probability if needed.

Example: Predict savings from reduced staff = \$10,000/year.

3) Determine Cash Flow - Project inflows and outflows over 3-5 years.

Example: Yearly benefits vs yearly costs shown in a cash flow table.

4) Assess Project's Economic Value - Use ROI, Break-Even Point (BEP), and ~~AE~~ Net Present value (NPV).

Example: If NPV = \$68,292 (positive), project is economically viable.

3a) Project Management

Project management is the discipline of initiating, planning, executing, controlling, and closing the work of a team to achieve specific goals and meet success criteria.

Execution Phase of Project Management

In this phase, the baseline project plan is put into action. Key activities include:

- 1) Executing the baseline plan - start project activities, assign resources, and ensure quality.
- 2) Monitoring progress - Track progress against the plan and adjust if needed.
- 3) Managing changes - Only approved changes are made and update in the plan.
- 4) Maintaining project workbook - Keep records of all project events and decisions.
- 5) Communicating project status - Update team members, managers, and customers regularly.

3) b) i) Cost-Benefit Analysis

It is a part of economic feasibility used to compare a project's costs and benefits over time. It helps determine whether the benefits justify the costs of the system.

Metrics to Measure Economic Value

1) Return on Investment (ROI): $\text{Net benefits} \div \text{Total costs}$.
Higher ROI = more profitable project.

2) Break-Even Point (BEP): Time required to recover the initial investment from net cash flows.

3) Discounted Cash Flow (DCF): Considers time value of money by converting future cash flows into present value.

4) Net Present Value (NPV): Difference between present value of benefits and present value of costs.

Positive NPV = project is acceptable.

3)a) Significance of Feasibility Analysis

Feasibility analysis guides whether a project should proceed. It identifies risks, assesses benefits vs. costs, and ensures the project is practical and achievable.

Phase of Project Management

Feasibility analysis is performed in the Project Initiation Phase.

Three Major Areas of Feasibility Study

- 1) Technical Feasibility - Checks whether current hardware, software, and technical resources can support the proposed system.
- 2) Economic Feasibility - Evaluates costs and benefits to see if the project is financially worthwhile.
- 3) Operational Feasibility - Examines whether the system will solve problems effectively and if users and management will support it.